PROJECT SYNOPSIS

PETROCHEMICAL TRADE ANALYTICS & PREDICTIVE INTELLIGENCE

Project Title:	Petrochemical Trade Analytics & Predictive Intelligence
Curriculum / Track:	HCL Career Shaper COE - Data Science & Advanced Analytics Track
Affiliated Program:	NASSCOM FutureSkills Prime Emerging Technologies Initiative
Candidate Name:	Anshu Kumar
Data & Horizon:	2015 – 2024 (10 Years) | UN Comtrade Benchmark ($302.18B Gross Trade)

1. Executive Summary & Abstract

This capstone project implements an end-to-end data analytics, econometric inference, and predictive machine learning platform evaluating international petrochemical trade flows. Tracking official UN Comtrade transactional datasets across eight major economies (China, Germany, India, Japan, Netherlands, Republic of Korea, Saudi Arabia, and the United States) from 2015 through 2024, the study analyzes the five primary building block feedstocks of global manufacturing: Ethylene (HS 290121), Propylene (HS 290122), Butadiene (HS 290124), Benzene (HS 290220), and Toluene (HS 290230). Total trade cleared reached 334.13 Million Metric Tons and $302.18 Billion USD. The platform incorporates Python ETL engineering, parametric hypothesis testing (Welch's t-test p = 1.42e-6, ANOVA p = 2.70e-9), Gradient Boosting regression (R^2 = 0.9543), an enterprise SQLite Star Schema, PySpark distributed pipelines, and a reactive Streamlit web dashboard with interactive choropleth maps and bilateral Sankey corridors.

2. Problem Statement & Motivation

Petrochemical feedstocks drive essential manufacturing across packaging, automotive components, construction materials, and pharmaceuticals. However, international trade suffers from acute price volatility tied to crude oil swings, refinery margin fluctuations, and regional supply deficits. Traditional trade management relies on fragmented, retrospective spreadsheets that fail to provide predictive foresight. This project solves this by unifying ETL pipelines, statistical rigor, predictive modeling, and business intelligence dashboards into a unified decision platform.

3. Core Objectives

• Data Pipeline: Design automated cleaning, imputation, currency harmonization, and temporal feature engineering.
• Statistical Rigor: Perform Welch's t-test and One-way ANOVA to evaluate pandemic structural breaks and cross-commodity price variance.
• Machine Learning: Build Gradient Boosting, Random Forest, and Ridge regression models alongside K-Means corridor clustering.
• Data Warehousing: Construct an SQLite Star Schema with Fact and Dimension tables and analytical window functions.
• Distributed Big Data: Develop PySpark pipelines for large-scale transactional trade aggregation.
• Business Intelligence: Deploy an interactive Streamlit dashboard with dual Dark/Light mode theming and scenario simulation.

4. Key Empirical Findings & Model Results

• Benchmark Market Scope: Cumulative Gross Trade of $302.18B USD (Exports: $148.44B, Imports: $153.74B, Balance: -$5.29B, Volume: 334.13 Mt).
• Chemical Family Split: Light Olefins account for 51.2% ($154.67B) and Aromatics account for 48.8% ($147.51B) of international clearing value.
• Statistical Validation: Structural shift from COVID-19 validated via Welch's t-test (t = -4.858, p = 1.42 x 10^-6, Reject H0). ANOVA confirms inter-commodity pricing divergence (F = 11.23, p = 2.70 x 10^-9).
• Predictive AI Benchmark: Gradient Boosting Regressor achieved R^2 = 0.9543, MAE = $19.45M, and RMSE = $34.12M, superior to Random Forest (R^2 = 0.9471).

5. Technology Stack Specifications

Core Languages & Frameworks:	Python 3.13, PySpark 3.5.0, SQLite3, Streamlit, DAX
Machine Learning & Stats:	Scikit-Learn, SciPy Stats, NumPy, Pandas, Joblib
Visualization & BI:	Plotly Express, Matplotlib, Seaborn, Power BI Desktop
Document & Model Outputs:	python-docx, ReportLab PDF, OpenPyXL Excel Financial Model

6. Academic & Corporate Conclusion

The project delivers a fully verified capstone solution complying with all HCL Career Shaper COE and NASSCOM evaluation criteria. All analytical models, database schemas, and interactive web tools have been compiled and packaged for instant assessment.